Supplementary Material for "DistilSCD: Output Distillation of a Remote-Sensing Foundation Model into a Compact Semantic Change Detector"

Mathis Magron, Eric Hervet, and Andy Couturier

All numbers below are generated from the run logs by the scripts in `paper/scripts/` of the code repository; SeK, Fscd and mIoU are in percentage points.

S1. SECOND: Per-Seed Results and the 200-Epoch Extension

Table SI lists every SECOND test run. Arms with the same seed share their initialization, data order and augmentation draws; their losses at the first optimizer step are identical, which we checked for every arm. Without distillation, the best epoch falls between 47 and 100 and the SeK then declines under the constant learning rate; with distillation, the best epoch is at or near the end of the 120-epoch budget. Because the distilled runs were still improving, we resumed them from epoch 120 to 200 after seeing the curves, a decision we report as post hoc. The extension adds +0.12 [0.05, 0.19] points to the best-epoch SeK (paired, $p = 0.008$), reaching 26.32 ± 0.11, i.e., +0.22 [+0.08, +0.35] above the teacher ($p = 0.01$); the curves are flat after epoch 150. Over the same 200 epochs, the undistilled student's best epoch is unchanged and its final SeK falls to 23.30; the distillation gain at 200 epochs is +2.42 on the best epoch and +2.96 on the final epoch.

S2. Screening Runs

Table SII lists the 32 screening runs (16 arms, two seeds) on the internal validation split. The split (2 671 training and 297 validation pairs from the official training set) is fixed and hashed and is released with the code. The decision rule ("best mean validation SeK; an increment counts if it exceeds 0.36 points") was written in the versioned lab journal before each of the three waves (waves 1–3 were trained at commits `2f8ece0`, `51205c5` and `8de169a`, the feature wave at `945879a`; see `documentation/distillation.md`). It is a record specified in advance, not a public registration. The change-weight grid was extended twice, from $\{0.5, 1, 2\}$ to $\{4, 8\}$ and then to 16, because the best value lay at its edge. Two caveats apply to the screening only. First, the SECOND teacher was trained by its authors on all 2 968 training pairs, including our 297 validation pairs, which can favor distillation over the ground-truth controls on validation; it is unlikely to explain the whole gap of about 1.8 points between distillation and the controls at the same weight. These screening controls were not run on the SECOND test set, so the SECOND test confirms the distillation gain, not the control comparison; a ground-truth-only control at the retained configuration was run instead on the location-disjoint Landsat-SCD folds, whose test tiles no teacher saw in training (Section VII of the main text). Second, two seeds detect only large increments; the screening ranks configurations, and the test confirmation with five seeds supports the conclusions.

S3. Teachers

A. Reproduction of the Published SECOND Checkpoints

Table SIII reports the two published teachers re-evaluated with our code. For the ViT-G teacher, the authors' evaluation code and ours agree to 10^{-8} on the same predictions; the deformable-attention CUDA operator and its PyTorch reference agree on 99.9994 % of the arg-max decisions; our SECOND copy and the authors' pre-processed copy are equivalent for evaluation (no pixel where the change map and the semantic labels disagree, among 444 M pixels, and identical class order). Half precision changes the ViT-G score by less than 0.01 point, so the cache was computed in float16. The VMamba-B checkpoint is sensitive to the input normalization: with the PerA statistics used by the released evaluation code it scores 23.73, and with ImageNet statistics it reproduces exactly the score stored in the checkpoint; its float16 inference is also degraded, so its cache was computed in float32 and stored in float16. We verified that the selective-scan kernels were not the cause (compiled kernel, PyTorch reference scan, and TF32 disabled give the same SeK on a 256-pair subset).

TABLE SI
SECOND TEST RESULTS PER SEED.

Arm	Epochs	Seed	SeK max	Best ep.	SeK final	Fscd	mIoU
change KD (ViT-G)	120	1	25.61	118	25.61	65.64	74.37
change KD (ViT-G)	120	2	25.50	115	25.47	65.53	74.33
change KD (ViT-G)	120	3	25.41	119	25.41	65.44	74.31
change KD (ViT-G)	120	4	25.69	117	25.68	65.72	74.43
change KD (ViT-G)	120	5	25.74	114	25.70	65.77	74.42
DistilSCD (ViT-G)	120	1	26.39	119	26.39	66.47	74.64
DistilSCD (ViT-G)	120	2	26.18	108	26.16	66.23	74.59
DistilSCD (ViT-G)	120	3	26.11	119	26.11	66.23	74.52
DistilSCD (ViT-G)	120	4	26.14	116	26.13	66.23	74.55
DistilSCD (ViT-G)	120	5	26.19	119	26.19	66.28	74.54
DistilSCD (VMamba-B)	120	1	25.69	118	25.68	65.74	74.37
DistilSCD (VMamba-B)	120	2	25.59	119	25.59	65.62	74.34
DistilSCD (VMamba-B)	120	3	25.46	105	25.45	65.56	74.25
DistilSCD (VMamba-B)	120	4	25.90	118	25.89	65.92	74.47
DistilSCD (VMamba-B)	120	5	25.82	119	25.82	65.78	74.46
no KD	120	1	23.87	100	23.73	64.45	73.18
no KD	120	2	23.77	94	23.61	64.35	73.14
no KD	120	3	23.91	62	23.48	64.45	73.20
no KD	120	4	23.94	66	23.55	64.39	73.31
no KD	120	5	23.76	47	23.37	64.19	73.25
no KD	120	6	24.20	62	24.04	64.72	73.36
no KD	120	42	23.84	72	23.73	64.38	73.18
DistilSCD (ViT-G)	200	1	26.45	182	26.39	–	–
DistilSCD (ViT-G)	200	2	26.36	151	26.28	–	–
DistilSCD (ViT-G)	200	3	26.19	143	26.08	–	–
DistilSCD (ViT-G)	200	4	26.25	196	26.24	–	–
DistilSCD (ViT-G)	200	5	26.37	191	26.35	–	–
no KD	200	1	23.87	100	23.26	64.45	73.18
no KD	200	2	23.77	94	23.23	64.35	73.14
no KD	200	3	23.91	62	23.24	64.45	73.20
no KD	200	4	23.94	66	23.28	64.39	73.31
no KD	200	5	23.76	47	23.33	64.19	73.25
no KD	200	6	24.20	62	23.52	64.72	73.37
no KD	200	42	23.84	72	23.28	64.38	73.18

Fscd and mIoU at the best epoch. The 200-epoch DistilSCD rows resume the 120-epoch runs (post-hoc extension); the 200-epoch no-KD rows are the full runs whose first 120 epochs give the 120-epoch rows.

B. Teacher Equivariance and the D_4 Cache

Between $T(g \cdot x)$ and $g \cdot T(x)$, the ViT-G teacher's change decision differs on 2.2–2.5 % of the pixels of the SECOND training set and its class decision on 2.7–3.0 % of the changed pixels. With about 20 % of pixels changed, a single-view cache would have given change targets inconsistent with the student's augmentation on a number of pixels equal to roughly a tenth of the changed area. On its own training set, read back from the half-precision cache, the teacher scores 33.24 SeK (Fscd 71.50, mIoU 77.77): the soft targets are far from a copy of the labels.

C. Landsat-SCD Fold Teachers

Table SIV summarizes the four retrained ViT-G teachers. They follow the authors' recipe (SGD, learning rate 0.1, momentum 0.9, polynomial decay, 10 % warm-up, 50 epochs, batch 4 with 2 accumulation steps, float16 autocast, gradient clipping 1.5, drop-path 0.3, D_4 and photometric augmentation), with two declared differences: the checkpoint is selected on the fold's validation SeK instead of the test Fscd, and activation checkpointing of the 40 ViT blocks is used to fit a 40 GB A100. On CPU with a ViT-B and drop-path 0.3, checkpointing gives gradients identical to the reference (maximum difference 0.0). The 486 PerA pre-trained tensors are loaded and checked before training.

TABLE SII
SCREENING RUNS ON THE SECOND INTERNAL VALIDATION SPLIT (2 671 / 297 PAIRS).

Arm tag	Seed	SeK max	Best ep.	SeK final	A100-h	GB
ctl-gt-l2	1	23.60	76	22.87	4.7	4.2
ctl-gt-l2	2	23.69	51	22.68	4.7	4.2
ctl-gtsmooth-l2	1	23.60	114	23.55	4.7	4.2
ctl-gtsmooth-l2	2	23.59	33	22.57	4.8	4.2
kdchg-l05	1	23.82	119	23.82	5.0	4.2
kdchg-l05	2	24.58	75	24.20	4.7	4.2
kdchg-l1	1	24.54	119	24.54	4.8	4.2
kdchg-l1	2	25.19	91	24.78	5.0	4.2
kdchg-l16	1	25.88	108	25.80	4.7	4.2
kdchg-l16	2	25.87	110	25.84	4.7	4.2
kdchg-l2	1	25.30	118	25.29	4.6	4.2
kdchg-l2	2	25.59	95	25.33	4.7	4.2
kdchg-l4	1	25.29	86	25.27	4.7	4.2
kdchg-l4	2	25.77	82	25.70	4.7	4.2
kdchg-l8	1	26.23	109	26.10	4.7	4.2
kdchg-l8	2	25.66	85	25.48	4.7	4.2
kdfeat-s12-l1	1	26.87	117	26.84	11.4	17.4
kdfeat-s12-l1	2	26.65	85	26.49	11.0	17.4
kdfeat-s12-l4	1	26.39	116	26.39	10.7	17.4
kdfeat-s12-l4	2	26.19	114	26.19	10.8	17.4
kdfeat-s123-l1	1	26.73	119	26.73	10.7	17.4
kdfeat-s123-l1	2	27.01	105	26.91	10.7	17.4
kdsem-l8-t2-all	1	26.79	115	26.75	4.7	4.2
kdsem-l8-t2-all	2	26.70	105	26.57	4.6	4.2
kdsem-l8-t2-changed	1	26.61	114	26.60	4.7	4.2
kdsem-l8-t2-changed	2	26.78	91	26.78	4.7	4.2
kdsem-l8-t4-all	1	26.27	114	26.27	4.7	4.2
kdsem-l8-t4-all	2	26.52	107	26.30	4.7	4.2
kdsem-l8-t4-changed	1	26.68	119	26.68	4.7	4.2
kdsem-l8-t4-changed	2	26.25	118	26.23	4.6	4.2
witness	1	23.40	66	23.07	4.7	4.2
witness	2	23.82	93	23.44	5.0	4.2

Tags: kdchg-lλ = change KD with weight λ; ctl-gt / ctl-gtsmooth = ground-truth controls; kdsem-l8-tT-mask = semantic KD; kdfeat-s(stages)-lλ = feature KD; witness = no KD.

TABLE SIII
RE-EVALUATION OF THE PUBLISHED SECOND TEACHERS ON THE FULL TEST SET (1 694 PAIRS).

Teacher	Setting	Normalization	SeK	Fscd	mIoU	ms (b8)
PerASCD ViT-G	precision fp32	PerA	26.108	66.410	74.331	1613.4
PerASCD ViT-G	precision fp16	PerA	26.108	66.411	74.331	489.7
PerASCD ViT-G	precision bf16	PerA	26.105	66.409	74.330	562.6
VMamba-B	precision fp32	imagenet	25.314	65.614	74.014	504.6
VMamba-B	precision fp16	imagenet	23.659	64.419	72.956	182.4
VMamba-B	precision bf16	imagenet	25.312	65.615	74.013	187.4
VMamba-B	normalisation, fp32	PerA (released evaluation code)	23.732	–	–	–
VMamba-B	normalisation, fp32	ImageNet	25.314	–	–	–
VMamba-B	normalisation, fp32	dataset statistics (commented in released code)	24.528	–	–	–
VMamba-B	normalisation, fp32	per-date 0-255	24.469	–	–	–

Published: ViT-G 26.11 (training log 26.1087); VMamba-B 25.31 (stored in the checkpoint: 0.25314). Latency at batch 8 on one A100 (the VMamba-B job ran on a different node).

S4. LANDSAT-SCD PROTOCOL

A. Overlap Verification

`scripts/verify_landsat_leakage.py` hashes the pixels of every image slot of the released LandsatSCD512 archive (two images per pair), independently of file names. It reports the number of distinct images, the

TABLE SIV
RETRAINED PERASCD VIT-G TEACHERS, ONE PER LANDSAT-SCD FOLD (50 EPOCHS, AUTHORS' RECIPE).

Fold	Test tiles	Val	Ep.	Gate	Pass	Test	Test final	Test max	A100-h	GB
0	00, 02, 20	21.85	28	22.50	no	28.61	29.12	29.99	6.4	28.5
1	01, 21, 30	25.74	42	23.36	yes	26.02	25.47	26.35	6.4	21.0
2	10, 12, 22	16.89	31	14.90	yes	22.48	23.00	23.10	6.7	23.5
3	11, 31, 32	23.25	11	20.70	yes	20.18	20.19	20.74	6.4	23.5

Val/Test: SeK at the best-validation epoch; Gate: mean best validation SeK of the three no-KD students of the fold; Test max is shown for information only (never used for selection).

TABLE SV
BUDGET PILOT ON LANDSAT-SCD FOLD 0 (STUDENT WITHOUT KD, ONE SEED).

Epochs	Val mean	Val slope	Test mean	Test slope
0-49	18.56	3.576	25.19	3.772
50-99	22.84	0.044	29.25	-0.005
100-149	22.63	-0.050	29.12	0.006
150-199	22.49	-0.007	29.12	-0.023
200-249	22.58	-0.008	29.01	-0.046
250-299	22.49	-0.011	28.77	-0.024
300-349	22.36	-0.032	28.76	-0.004
350-399	22.16	-0.071	28.71	-0.004
400-449	22.11	-0.004	28.71	-0.002

Slopes in SeK points per 10 epochs. Rule fixed before the run, applied to validation only: first 50-epoch window with a validation slope below 0.05, i.e. epochs 50–99, hence a 100-epoch budget.

pairs of each split whose two images exist identically in the training set, the consistency of the class labels across duplicates, the hypothesis that the tile index of the name identifies a location (every (index, year) group holds a single distinct image, and no image is shared by two indices in the same year), and the SeK of a lookup oracle that labels each test pixel with the classes annotated for the same image in the training set. Table II of the paper reports the results. The same hash check finds no exact duplicate between the SECOND training (5 936 images) and test (3 388 images) sets; partial spatial overlap without exact duplicates is not tested.

B. Fold Construction and Budget

The folds (`splits/LandsatSCD_loc/fold{0..3}`) result from an exhaustive search over the 15 400 partitions of the 12 tiles into four groups of three, scored on the balance of the nine transition types and of the change ratio; in each fold, one non-test tile serves as validation. Table SV reports the budget pilot on fold 0: the rule fixed before the run, read on validation only, selects the window of epochs 50–99, hence the 100-epoch budget used for every arm and fold.

C. All Runs

Table SVI lists the 24 Landsat-SCD runs. Fold-0 distilled runs read the cache `perascd_landsat_f0_train_exploratory`, written only because the gate was lifted explicitly; the lift is recorded in the cache directory and in each run's configuration.

S5. STUDENT DESIGN BEFORE THIS STUDY

The student was designed earlier by the first author as an efficient SCD model and its recipe was frozen before the distillation study. Table SVII summarizes the design results that fixed the recipe, for transparency. Three blocks inherited from that design (the checkerboard state-space fusion, the frequency branch and the change-guided attention) have no detectable effect; the fusion block was removed, the other two remain in the network that is distilled here. These design experiments selected epochs on the SECOND test set, as is customary on this dataset; their effects were required to hold on both the best and the final epoch.

TABLE SVI
LANDSAT-SCD LOCATION-DISJOINT FOLDS, ALL RUNS.

Fold	Arm	Seed	Val max	Best ep.	Test	Fscd	mIoU	Test final	Val slope
0	no KD	1	22.77	61	29.07	72.07	75.73	28.88	0.009
0	no KD	2	22.47	97	28.35	71.60	75.33	28.34	0.109
0	no KD	3	22.27	99	28.84	71.99	75.58	28.84	0.089
0	DistilSCD	1	24.41	77	31.19	72.83	76.81	31.24	-0.044
0	DistilSCD	2	24.08	82	31.31	72.99	76.84	31.45	0.020
0	DistilSCD	3	24.68	99	31.13	72.78	76.84	31.13	0.064
1	no KD	1	23.44	98	25.58	69.43	74.57	25.59	0.113
1	no KD	2	23.39	44	25.65	69.47	74.61	25.79	-0.053
1	no KD	3	23.25	35	25.49	69.29	74.53	25.76	0.043
1	DistilSCD	1	25.72	70	27.93	70.14	75.97	28.11	-0.035
1	DistilSCD	2	25.52	99	28.00	70.25	76.04	28.00	0.021
1	DistilSCD	3	25.53	98	28.22	70.43	76.13	28.22	0.030
2	no KD	1	15.27	74	21.74	68.52	73.09	21.61	-0.052
2	no KD	2	14.56	98	21.12	68.03	72.72	21.11	0.143
2	no KD	3	14.87	84	21.58	68.46	72.99	21.61	0.012
2	DistilSCD	1	18.73	93	23.41	69.23	73.95	23.39	0.097
2	DistilSCD	2	17.52	69	23.28	69.06	73.88	23.38	0.023
2	DistilSCD	3	17.49	90	23.27	69.11	73.89	23.31	0.079
3	no KD	1	20.84	60	18.23	63.88	71.18	18.57	-0.024
3	no KD	2	21.06	89	19.09	64.65	71.65	19.10	0.091
3	no KD	3	20.21	57	17.94	63.64	71.09	18.06	-0.008
3	DistilSCD	1	23.97	97	21.56	66.32	73.15	21.57	0.027
3	DistilSCD	2	24.14	97	21.89	66.26	73.36	21.87	0.038
3	DistilSCD	3	23.55	98	21.64	66.24	73.16	21.64	0.032

Test, Fscd, mIoU at the best-validation epoch; val slope in SeK points per 10 epochs over epochs 70–99. Fold 0 is exploratory (teacher gate lifted).

TABLE SVII
STUDENT DESIGN RESULTS OBTAINED BEFORE THIS STUDY (SECOND, SEK POINTS).

Component	Contribution	95 % CI	Verdict
C2S2 block (checkerboard + MCA-SF + S6 scan), removed	0.16	[-0.04, 0.36]	not detectable
MCA-SF alone	0.15	[-0.24, 0.52]	not detectable
change-guided attention (CGA), kept	-0.01	–	not detectable
FFT frequency branch, kept	-0.46	–	not established
wider decoder (+3.5 M, +12 GMACs)	-0.07	–	not detectable
removing the SeK loss term of Mamba-FCS	1.25	–	established
temporal-swap augmentation	0.96	–	established
EMA of weights	0.74	–	established (0.74-0.87)
DySample instead of bilinear upsampling	0.75	–	established

From the design campaign of the student (about 150 runs, 4–8 seeds per configuration, SECOND test with best-epoch and final-epoch readings). These results fixed the student recipe that is then frozen in this paper; they are reported for transparency and are not part of the distillation evidence.

S6. CONTROL EXPERIMENTS

Table SVIII lists every control run of Section VII of the main text, and Table SIX the checks of the student-as-teacher caches used for self-distillation. The caches have the format of the teacher caches (eight D_4 views, $2C + 1$ logit channels at one quarter of the tile resolution) and are read by the same data loader. Pooling the student's full-resolution logits to that resolution costs 0.22 SeK points on SECOND but 2.7–3.4 on the Landsat-SCD folds, whose changed regions are small and fragmented. The analysis script is `paper/scripts/analyze_revision_a.py`; the plan, written before the runs were evaluated, is `documentation/revision_vague_a.md`.

S7. COMPUTE

Table SX lists the GPU time used by this study, about 553 A100-hours in total, excluding teacher caches and checks (about 10 A100-hours).

TABLE SVIII
CONTROL RUNS (SECTION VII OF THE MAIN TEXT), PER RUN.

Dataset	Fold	Arm	Seed	SeK	Ep.	SeK final	ε
Landsat-SCD	1	ground truth only	1	25.40	88	25.39	0.21
Landsat-SCD	1	ground truth only	2	25.72	37	25.59	0.21
Landsat-SCD	1	ground truth only	3	25.53	71	25.53	0.21
Landsat-SCD	2	ground truth only	1	21.02	65	21.03	0.25
Landsat-SCD	2	ground truth only	2	20.74	47	20.84	0.25
Landsat-SCD	2	ground truth only	3	20.89	65	20.82	0.25
Landsat-SCD	3	ground truth only	1	18.26	85	18.32	0.28
Landsat-SCD	3	ground truth only	2	18.46	95	18.46	0.28
Landsat-SCD	3	ground truth only	3	18.47	84	18.48	0.28
Landsat-SCD	1	self-distillation	2	25.39	99	25.39	–
Landsat-SCD	1	self-distillation	3	25.46	98	25.49	–
Landsat-SCD	2	self-distillation	2	21.01	89	21.06	–
Landsat-SCD	2	self-distillation	3	21.09	99	21.09	–
Landsat-SCD	3	self-distillation	2	18.79	89	18.86	–
Landsat-SCD	3	self-distillation	3	18.89	95	18.91	–
SECOND	–	no KD, cosine LR	1	23.66	35	22.95	–
SECOND	–	no KD, cosine LR	2	23.48	37	22.68	–
SECOND	–	no KD, cosine LR	3	23.42	57	22.75	–
SECOND	–	self-distillation	2	24.66	103	24.65	–
SECOND	–	self-distillation	3	24.52	99	24.46	–
SECOND	–	self-distillation	4	24.68	118	24.68	–

SECOND: best test epoch over 120 epochs; Landsat-SCD: test SeK at the best-validation epoch over 100 epochs. ε: label-smoothing parameter of the ground-truth-only control, matched to the fold teacher (scripts/kd_control_eps.py).

TABLE SIX
STUDENT-AS-TEACHER CACHES FOR THE SELF-DISTILLATION CONTROL.

Cache	Pairs	SeK full	SeK cache	Fscd full	Fscd cache	D4 agr. (%)
student_landsat_f1_train	1590	54.05	50.64	86.55	85.18	98.0
student_landsat_f2_train	1590	52.09	49.43	85.25	84.16	97.9
student_landsat_f3_train	1577	50.21	47.44	84.83	83.69	97.8
student_second_train	2968	53.54	53.32	84.32	84.21	97.8

Training-set scores of the undistilled seed-1 student: at full resolution (identity view) and from the cached targets (4×4 average-pooled logits, bilinearly upsampled). D4 agr.: minimum over the seven non-identity views of the change agreement between the cached view and the transformed identity view.

TABLE SX
COMPUTE USED IN THIS STUDY.

Study	Runs	A100-h	Source
SECOND: KD screening, 3 logit waves (26 runs)	26	126.8	sacct
SECOND: feature-KD screening (6 runs)	6	66.4	sacct
SECOND: test confirmation, ViT-G teacher (10 runs)	10	66.0	sacct
SECOND: end-to-end trial	1	0.5	sacct
SECOND: VMamba-B teacher study (5 runs + teacher eval/cache)	5	32.6	sacct
SECOND: 200-epoch extension (5 resumes)	5	21.7	sacct
Landsat random split: student baseline, 600 epochs (5 runs, incl. first 250)	5	70.5	timing.csv
Landsat fold 0: 600-epoch budget pilot (stopped at epoch 477)	1	11.2	timing.csv
Landsat folds: student baselines (12 runs)	12	28.2	timing.csv
Landsat folds: PerASCD ViT-G teachers (4 runs)	4	25.8	timing.csv
Landsat folds: KD students (12 runs)	12	29.0	timing.csv
Revision controls: self-distillation, ground-truth targets, cosine LR (21 runs)	21	74.3	timing.csv
TOTAL (excluding teacher caches and checks, about 10 A100-h)	108	552.9	sum

S8. REPRODUCIBILITY

The repository contains the training and evaluation code (`distilscd/`), the scripts for teacher evaluation and caching (`scripts/teacher/`), the SLURM launchers used on the Narval cluster (`scripts/train_kd_second.sbatch`, `scripts/train_kd_landsat.sbatch`, `scripts/teacher/teacher_landsat.sbatch`), the splits (`splits/`), the per-run logs used for every table, and the scripts that regenerate the tables, numbers and figures of the paper from those logs (`paper/scripts/build_tables.py`, `make_numbers.py`, `make_tables_tex.py`, `make_figures.py`). Each run directory stores the full configuration and the commit it was trained with; a launcher refuses to reuse a run directory for a different configuration. GPU training is not bit-reproducible, so replays differ by up to 0.5 SeK on Landsat-SCD; the path without distillation was verified to be bit-identical on CPU before and after the distillation code was added.